

Resultater fra QH4 for Hilde Celine Løvland

Resultat for dette forsøket: 0 av 11

Innlevert 24. sep. i 16.26

Dette forsøket tok 71 minutter

Feil svar



Spørsmål 1

0 / 5 poeng

Hydraulic system . Consider a variable displacement pump set at a constant pressure supply and connected to a base unit, see Figure Q4.8 . The base unit is a 4/3-way proportional DCV, a counterbalance valve, and an extracting cylinder, see Figure Q4.4 . Pump max displacement (D_{Max}) = 23 cm³/rev . Pump speed (n) = 1120 rev/min . Pump volumetric efficiency (η_{vP}) = 0.92 . Main crack pressure (p_{crM}) = 240 bar . The 4/3-way proportional directional control valve has identical discharge coefficients: $C_{dPA} = C_{dBT} = C_{dPB} = C_{dAT} = C_d$. The 4/3-way proportional directional control valve has symmetrical discharge areas: $A_{dMax,PA} = A_{dMax,PB}$ and $A_{dMax,BT} = A_{dMax,AT}$. Liquid density (ρ) = 875 kg/m³ . Discharge coefficient (C_d) = 0.70 . Maximum discharge area ($A_{dMax,PA}$) = 18 mm² . Maximum discharge area ($A_{dMax,BT}$) = 18 mm² . Max valve spool travel (x_{Max}) = 10.0 mm . Valve spool travel (x) = 7.5 mm . Pilot area ratio (α) = 1.7 . Crack pressure (p_{cr}) = 240 bar . The cylinder is extracting . The cylinder is subjected to an assistive (negative) load . Cylinder diameter (d) = 50 mm . Cylinder rod diameter (d_R) = 25 mm . Cylinder hydromechanical efficiency (η_{hmC}) = 0.92 . Cylinder load force (F_L) = 34.0 kN . Pump set pressure (p_{set}) = 19 bar . What is the pressure (p_{Bu}) in bar?

225,3044

229,9111 (med marginen: 2,2991)

Feil svar



Spørsmål 2

0 / 1 poeng

Valve bandwidth requirement . Consider a motor connected to a servo valve, see Figure Q4.10 . The two hydraulic lines are identical, i.e., $V_{L1} = V_{L2}$ and $V_L = V_{L1} + V_{L2}$. Bulk modulus (β) = 1382 MPa . Motor displacement (D) = 160 cm³/rev . Total line volume (V_L) = 0.81 l . Mass moment of inertia (J) = 2.70 kg*m² . What is the minimum required valve bandwidth ($f_{v@90deg}$) in Hz?

36,9961

17,6644 (med marginen: 0,1766)

Feil svar



Spørsmål 3

0 / 2 poeng

Minimum rated flow for servo valve . Consider a motor connected to a servo valve, see Figure Q4.12 . Motor displacement (D) = 500 cm³/rev . Mass moment of inertia (J) = 15.20 kg*m² . Supply pressure (p_s) = 250 bar . Payload total rotation ($\Delta\theta$) = 81.5 rad . Total task time (Δt) = 5.258 s . Ramp time (t_R) = 1.202 s . External moment (M_1) = 97 . External moment (M_2) = 251 . External moment (M_3) = 661 . Consider a specific time during the task (t) = 0.868 s . What is the no-load flow (Q_{NL}) at the considered time (t) in l/min?

79,1706

76,3448 (med marginen: 0,7634)

Feil svar



Spørsmål 4

0 / 3 poeng

No-load flow in servo systems . Consider a symmetrical cylinder connected to a servo valve, see Figure Q4.11 . Cylinder diameter (d) = 40 mm . Cylinder rod diameter (d_R) = 20 mm . Mass (m) = 961 kg . Supply pressure (p_s) = 250 bar . Valve rated pressure (p_r) = 10 bar . Payload total travel distance (Δs) = 384 mm . Total task time (Δt) = 0.694 s . Ramp time (t_R) = 0.143 s . External force (F_1) = 10.6 kN . External force (F_2) = 6.65 kN . External force (F_3) = -12.16 kN . What is the minimum required valve flow ($Q_{r,min}$) in l/min?

65,8934

13,2973 (med marginen: 0,1329)

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